Quantum Mechanics II

Solutions to the Problems in the Midterm Exam

Spring, 2025

1. (25 points) Complete the following five problems (5 points each).

(1) A particle of spin 1/2 and magnetic moment $\hat{\vec{\mu}} = g\hat{\vec{S}}$ is in a magnetic field. With only the spin degree of freedom of the particle considered the Hamiltonian operator of the particle is given by $\hat{H} = \frac{A}{\sqrt{2}}(\hat{\sigma}_x + \hat{\sigma}_y)$, where A is a positive real constant and $\hat{\sigma}_x$ and $\hat{\sigma}_y$ are Pauli operators. At $t = 0$, the spin of the particle is in the state $|+\rangle$, $|\psi(0)\rangle = |+\rangle$. Evaluate the time evolution operator $\hat{U}(t,0) = e^{-i\hat{H}t/\hbar}$. Find the state vector for the spin of the particle at $t > 0$ by using the time evolution operator $\hat{U}(t,0)$. How long does it take for the spin of the particle to be in the state $|-\rangle$ for the first time after $t = 0$?

Formulas provided: $\hat{\vec{S}} = \frac{\hbar}{2}\hat{\vec{\sigma}}$, $\hat{S}_\pm = \hat{S}_x \pm i\hat{S}_y$, $\hat{S}_-|+\rangle = \hbar|-\rangle$, $\hat{S}_+|-\rangle = \hbar|+\rangle$, $\hat{\sigma}_x\hat{\sigma}_y + \hat{\sigma}_y\hat{\sigma}_x = 0$,

$$e^x = \sum_{n=0}^{\infty}\frac{x^n}{n!},\quad \cos x = \sum_{n=0}^{\infty}\frac{(-1)^n x^{2n}}{(2n)!},\quad \sin x = \sum_{n=0}^{\infty}\frac{(-1)^n x^{2n+1}}{(2n+1)!}.$$

We first note that

$$\begin{aligned}
(\hat{\sigma}_x + \hat{\sigma}_y)^2 &= \hat{\sigma}_x^2 + \hat{\sigma}_y^2 + \hat{\sigma}_x\hat{\sigma}_y + \hat{\sigma}_y\hat{\sigma}_x = 2,\\
(\hat{\sigma}_x + \hat{\sigma}_y)^3 &= 2(\hat{\sigma}_x + \hat{\sigma}_y),\\
(\hat{\sigma}_x + \hat{\sigma}_y)^4 &= 4,\\
(\hat{\sigma}_x + \hat{\sigma}_y)^5 &= 4(\hat{\sigma}_x + \hat{\sigma}_y),\\
&\vdots
\end{aligned}$$

Expanding $\hat{U}(t,0) = e^{-i\hat{H}t/\hbar}$ as a Taylor series, we have

$$\begin{aligned}
\hat{U}(t,0) = e^{-i\hat{H}t/\hbar} &= e^{-iA(\hat{\sigma}_x+\hat{\sigma}_y)t/\sqrt{2}\hbar}\\
&= \sum_{n=0}^{\infty}(-1)^n\left(\frac{-iAt}{\sqrt{2}\hbar}\right)^{2n}\frac{(\hat{\sigma}_x+\hat{\sigma}_y)^{2n}}{(2n)!} + \sum_{n=0}^{\infty}(-1)^n\left(\frac{-iAt}{\sqrt{2}\hbar}\right)^{2n+1}\frac{(\hat{\sigma}_x+\hat{\sigma}_y)^{2n+1}}{(2n+1)!}\\
&= \sum_{n=0}^{\infty}(-1)^n\left(\frac{-iAt}{\sqrt{2}\hbar}\right)^{2n}\frac{2^n}{(2n)!} + (\hat{\sigma}_x+\hat{\sigma}_y)\sum_{n=0}^{\infty}(-1)^n\left(\frac{-iAt}{\sqrt{2}\hbar}\right)^{2n+1}\frac{2^n}{(2n+1)!}\\
&= \sum_{n=0}^{\infty}\left(\frac{At}{\hbar}\right)^{2n}\frac{1}{(2n)!} - \frac{i(\hat{\sigma}_x+\hat{\sigma}_y)}{\sqrt{2}}\sum_{n=0}^{\infty}\left(\frac{At}{\hbar}\right)^{2n+1}\frac{1}{(2n+1)!}\\
&= \cos\big(At/\hbar\big) - \frac{i(\hat{\sigma}_x+\hat{\sigma}_y)}{\sqrt{2}}\sin\big(At/\hbar\big)\\
&= \cos\big(At/\hbar\big) - \frac{1}{\sqrt{2}\hbar}\big[\,(i+1)\hat{S}_+ \;+(i-1)\hat{S}_-\,\big]\sin\big(At/\hbar\big).
\end{aligned}$$

The state vector of the spin of the particle at time $t > 0$ is given by

$$\begin{aligned}
|\psi(t)\rangle = \hat{U}(t,0)\,|\psi(0)\rangle &= \left\{\cos\big(At/\hbar\big) - \frac{1}{\sqrt{2}\hbar}\big[\,(i+1)\hat{S}_+ \;+(i-1)\hat{S}_-\,\big]\sin\big(At/\hbar\big)\right\}|+\rangle\\
&= \cos\big(At/\hbar\big)\,|+\rangle \;-\frac{1}{\sqrt{2}}(i-1)\,|-\rangle\sin\big(At/\hbar\big)
\end{aligned}$$

which indicates that at instants

$$t_n = \frac{(2n+1)\pi\hbar}{2A},\quad n = 0, 1, 2, \cdots$$

determined from $\cos\big(At/\hbar\big) = 0$, the spin of the particle is in the state $|-\rangle$. Thus, the time it takes for the spin of the particle to be in the state $|-\rangle$ for the first time after $t = 0$ is $\dfrac{\pi\hbar}{2A}$.

(2) A transformation to the position operator $\hat{x}$ of the one-dimensional harmonic oscillator of mass m and angular frequency ω is given by $\hat{S}^\dagger(r)\hat{x}\hat{S}(r)$, where the operator $\hat{S}(r)$ is defined by $\hat{S}(r) = e^{r(\hat{a}^{\dagger 2} - \hat{a}^2)/2}$ with r a positive real number and $\hat{a}^\dagger$ and $\hat{a}$ respectively the particle creation and annihilation operators of the oscillator. Evaluate $\hat{S}^\dagger(r)\hat{x}\hat{S}(r)$.

Formulas provided: $\big[\,\hat{a},\,\hat{a}^\dagger\,\big] = 1,\ \big[\,\hat{a},\,\hat{a}\big] = \big[\,\hat{a}^\dagger,\,\hat{a}^\dagger\,\big] = 0,\ \hat{x} = \sqrt{\dfrac{\hbar}{2m\omega}}\,\big(\hat{a} + \hat{a}^\dagger\big),$

$$e^{\hat{A}}\hat{B}e^{-\hat{A}} = \hat{B} + \big[\hat{A},\hat{B}\big] + \frac{1}{2!}\big[\hat{A},\big[\hat{A},\hat{B}\big]\big] + \frac{1}{3!}\big[\,\hat{A},\big[\hat{A},\big[\,\hat{A},\hat{B}\big]\big]\big] + \cdots.$$

Let $\hat{A} = r(\hat{a}^2 - \hat{a}^{\dagger 2})/2$ and $\hat{B} = \hat{x} = \sqrt{\hbar/2m\omega}\,\big(\hat{a} + \hat{a}^\dagger\big)$. For $\big[\hat{A},\hat{B}\big]$, we have

$$\begin{aligned}\big[\hat{A},\hat{B}\big] &= \frac{r}{2}\sqrt{\frac{\hbar}{2m\omega}}\big[\,\hat{a}^2 - \hat{a}^{\dagger 2},\hat{a} + \hat{a}^\dagger\,\big] = \frac{r}{2}\sqrt{\frac{\hbar}{2m\omega}}\Big\{\big[\,\hat{a}^2,\hat{a}^\dagger\,\big] - \big[\,\hat{a}^{\dagger 2},\hat{a}\,\big]\Big\}\\ &= r\sqrt{\frac{\hbar}{2m\omega}}\,\big(\hat{a} + \hat{a}^\dagger\big) = r\hat{x} = r\hat{B}.\end{aligned}$$

For $\big[\hat{A},\big[\hat{A},\hat{B}\big]\big]$, we have

$$\big[\hat{A},\big[\hat{A},\hat{B}\big]\big] = r\big[\hat{A},\hat{B}\big] = r^2\hat{x} = r^2\hat{B}.$$

For $\big[\,\hat{A},\big[\hat{A},\big[\,\hat{A},\hat{B}\big]\big]\big]$, we have

$$\big[\,\hat{A},\big[\hat{A},\big[\,\hat{A},\hat{B}\big]\big]\big] = r^2\big[\hat{A},\hat{B}\big] = r^3\hat{x} = r^3\hat{B}.$$

We then have

$$\begin{aligned}\hat{S}^\dagger(r)\hat{x}\hat{S}(r) &= \hat{x} + r\hat{x} + \frac{1}{2!}r^2\hat{x} + \frac{1}{3!}r^3\hat{x} + \cdots\\ &= \left(1 + r + \frac{1}{2!}r^2 + \frac{1}{3!}r^3 + \cdots\right)\hat{x}\\ &= e^r\hat{x}.\end{aligned}$$

(3) Calculate the uncertainties Δx and Δp_x of the position and momentum operators, $\hat{x}$ and $\hat{p}_x$, of the one-dimensional harmonic oscillator of mass m and angular frequency ω in the normalized coherent state $|\alpha\rangle$ with α a complex number. How is the Heisenberg uncertainty relation satisfied?

Formulas provided: $\hat{x} = \sqrt{\dfrac{\hbar}{2m\omega}}\,\big(\hat{a} + \hat{a}^\dagger\big),\ \hat{p}_x = -i\sqrt{\dfrac{m\hbar\omega}{2}}\,\big(\hat{a} - \hat{a}^\dagger\big),\ \Delta A = \sqrt{\langle\hat{A}^2\rangle - \langle\hat{A}\rangle^2},\ \hat{a}\,|\alpha\rangle = \alpha\,|\alpha\rangle.$

In the following calculations, we will repeatedly use

$$\begin{aligned}&\hat{a}\,|\alpha\rangle = \alpha\,|\alpha\rangle,\\ &\langle\alpha|\,\hat{a}^\dagger = \langle\alpha|\,\alpha^*.\end{aligned}$$

For $\langle\hat{x}\rangle$, we have

$$\langle\hat{x}\rangle = \sqrt{\frac{\hbar}{2m\omega}}\,\langle\alpha|\hat{a} + \hat{a}^\dagger|\alpha\rangle = \sqrt{\frac{\hbar}{2m\omega}}\,\big(\alpha + \alpha^*\big)\langle\alpha|\alpha\rangle = \sqrt{\frac{\hbar}{2m\omega}}\,\big(\alpha + \alpha^*\big).$$

For $\langle \hat{x}^2 \rangle$, we have

$$\begin{aligned}
\langle \hat{x}^2 \rangle &= \frac{\hbar}{2m\omega} \, \langle \alpha | (\hat{a} + \hat{a}^\dagger)^2 | \alpha \rangle \\
&= \frac{\hbar}{2m\omega} \, \langle \alpha | \hat{a}^2 + \hat{a}^{\dagger 2} + \hat{a}\hat{a}^\dagger + \hat{a}^\dagger \hat{a} | \alpha \rangle \\
&= \frac{\hbar}{2m\omega} \, \langle \alpha | \hat{a}^2 + \hat{a}^{\dagger 2} + 1 + 2\hat{a}^\dagger \hat{a} | \alpha \rangle \\
&= \frac{\hbar}{2m\omega} \, \big(\alpha^2 + \alpha^{*2} + 1 + 2\alpha^* \alpha \big) \\
&= \frac{\hbar}{2m\omega} \, \big[\, (\alpha + \alpha^*)^2 + 1 \, \big].
\end{aligned}$$

We then have

$$\Delta x = \sqrt{\langle \hat{x}^2 \rangle - \langle \hat{x} \rangle^2} = \sqrt{\frac{\hbar}{2m\omega}} \sqrt{(\alpha + \alpha^*)^2 + 1 - (\alpha + \alpha^*)^2} = \sqrt{\frac{\hbar}{2m\omega}}.$$

For $\langle \hat{p}_x \rangle$, we have

$$\langle \hat{p}_x \rangle = -i \sqrt{\frac{m\hbar\omega}{2}} \, \langle \alpha | \hat{a} - \hat{a}^\dagger | \alpha \rangle = -i \sqrt{\frac{m\hbar\omega}{2}} \, \big(\alpha - \alpha^* \big).$$

For $\langle \hat{p}_x^2 \rangle$, we have

$$\begin{aligned}
\langle \hat{p}_x^2 \rangle &= -\frac{m\hbar\omega}{2} \, \langle \alpha | (\hat{a} - \hat{a}^\dagger)^2 | \alpha \rangle \\
&= -\frac{m\hbar\omega}{2} \, \langle \alpha | \hat{a}^2 + \hat{a}^{\dagger 2} - \hat{a}\hat{a}^\dagger - \hat{a}^\dagger \hat{a} | \alpha \rangle \\
&= -\frac{m\hbar\omega}{2} \, \langle \alpha | \hat{a}^2 + \hat{a}^{\dagger 2} - 1 - 2\hat{a}^\dagger \hat{a} | \alpha \rangle \\
&= -\frac{m\hbar\omega}{2} \, \big(\alpha^2 + \alpha^{*2} - 1 - 2\alpha^* \alpha \big) \\
&= \frac{m\hbar\omega}{2} \, \big[\, 1 - (\alpha - \alpha^*)^2 \, \big].
\end{aligned}$$

We then have

$$\Delta p_x = \sqrt{\langle \hat{p}_x^2 \rangle - \langle \hat{p} \rangle_x^2} = \sqrt{\frac{m\hbar\omega}{2}} \sqrt{1 - (\alpha - \alpha^*)^2 + (\alpha - \alpha^*)^2} = \sqrt{\frac{m\hbar\omega}{2}}.$$

For the product $\Delta x \Delta p_x$, we have

$$\Delta x \Delta p_x = \sqrt{\frac{\hbar}{2m\omega}} \times \sqrt{\frac{m\hbar\omega}{2}} = \frac{\hbar}{2}.$$

Thus, the Heisenberg uncertainty relation is satisfied with the minimum value $\dfrac{\hbar}{2}$ taken on the right hand side.

(4) Evaluate $e^{-i\pi \hat{L}_y / 2\hbar} \hat{L}_x e^{i\pi \hat{L}_y / 2\hbar}$, where $\hat{L}_x$ and $\hat{L}_y$ are the x- and y-component operators of the orbital angular momentum.

Formulas provided: $e^{\hat{A}} \hat{B} e^{-\hat{A}} = \hat{B} + \big[\hat{A}, \hat{B} \big] + \dfrac{1}{2!} \big[\hat{A}, \big[\hat{A}, \hat{B} \big] \big] + \dfrac{1}{3!} \big[\, \hat{A}, \big[\hat{A}, \big[\, \hat{A}, \hat{B} \big] \big] \big] + \cdots$,

$$\big[\, \hat{L}_\alpha, \hat{L}_\beta \, \big] = i\hbar \sum_\gamma \epsilon_{\alpha\beta\gamma} \hat{L}_\gamma, \; \cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}, \; \sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}.$$

For the evaluation of $e^{-i\pi\hat{L}_y/2\hbar}\hat{L}_x e^{i\pi\hat{L}_y/2\hbar}$, we set $\hat{A} = -i\pi\hat{L}_y/2\hbar$ and $\hat{B} = \hat{L}_x$. We have

$$\begin{aligned}
&\big[\,\hat{A},\hat{B}\,\big] = -\frac{i\pi}{2\hbar}\big[\,\hat{L}_y,\hat{L}_x\,\big] = -\frac{\pi}{2}\hat{L}_z,\\
&\big[\hat{A},\big[\hat{A},\hat{B}\big]\big] = \frac{i\pi^2}{4\hbar}\big[\,\hat{L}_y,\hat{L}_z\,\big] = -\frac{\pi^2}{4}\hat{L}_x,\\
&\big[\,\hat{A},\big[\hat{A},\big[\,\hat{A},\hat{B}\big]\big]\big] = \frac{i\pi^3}{8\hbar}\big[\,\hat{L}_y,\hat{L}_x\,\big] = \frac{\pi^3}{8}\hat{L}_z,\\
&\big[\,\hat{A},\big[\hat{A},\big[\,\hat{A},\big[\hat{A},\hat{B}\big]\big]\big]\big] = -\frac{i\pi^4}{16\hbar}\big[\,\hat{L}_y,\hat{L}_z\,\big] = \frac{\pi^4}{16}\hat{L}_x,\\
&\qquad\vdots
\end{aligned}$$

We then have

$$\begin{aligned}
e^{-i\pi\hat{L}_y/2\hbar}\hat{L}_x e^{i\pi\hat{L}_y/2\hbar} &= \hat{L}_x - \frac{\pi}{2}\hat{L}_z - \frac{1}{2!}\frac{\pi^2}{4}\hat{L}_x + \frac{1}{3!}\frac{\pi^3}{8}\hat{L}_z + \frac{1}{4!}\frac{\pi^4}{16}\hat{L}_x + \cdots\\
&= \left[\,1 - \frac{1}{2!}\left(\frac{\pi}{2}\right)^2 + \frac{1}{4!}\left(\frac{\pi}{2}\right)^4 + \cdots\right]\hat{L}_x - \left[\,\left(\frac{\pi}{2}\right) - \frac{1}{3!}\left(\frac{\pi}{2}\right)^3 + \cdots\right]\hat{L}_z\\
&= \cos(\pi/2)\hat{L}_x - \sin(\pi/2)\hat{L}_z = -\hat{L}_z.
\end{aligned}$$

(5) The Hamiltonian operator of a particle of mass μ and charge q moving in the presence of a uniform magnetic field $\vec{B} = B\vec{e}_z$ is given by $\hat{H} = \dfrac{1}{2\mu}\big[\,\hat{\vec{p}} - q\vec{A}(\vec{r}\,)\,\big]^2$. The velocity operator is given by $\hat{\vec{v}} = \dfrac{1}{\mu}\big[\,\hat{\vec{p}} - q\vec{A}(\vec{r}\,)\,\big]$. Derive the equations of motion for the component operators $\hat{v}_x$, $\hat{v}_y$, and $\hat{v}_z$ of the velocity of the particle by using the Heisenberg equation of motion. Solve the derived equations of motion under the initial conditions $\hat{v}_x(t=0) = \hat{v}_x(0)$, $\hat{v}_y(t=0) = \hat{v}_y(0)$, and $\hat{v}_z(t=0) = \hat{v}_z(0)$.

Formulas provided: $\big[\,\hat{v}_\alpha,\hat{v}_\beta\,\big] = \dfrac{i\hbar q}{\mu^2}\displaystyle\sum_\gamma \epsilon_{\alpha\beta\gamma}B_\gamma$, $i\hbar\dfrac{d}{dt}\hat{A}_H(t) = \big[\hat{A}_H(t),\hat{H}_H\big] + i\hbar\left(\dfrac{d}{dt}\hat{A}_S(t)\right)_H$.

We first evaluate the relevant commutators. For $\big[\,\hat{v}_\alpha,\hat{H}\,\big]$, we have

$$\begin{aligned}
\big[\,\hat{v}_\alpha,\hat{H}\,\big] &= \frac{1}{2}\mu\sum_\beta\big[\,\hat{v}_\alpha,\hat{v}_\beta^2\,\big] = \frac{1}{2}\mu\sum_\beta\Big(\big[\,\hat{v}_\alpha,\hat{v}_\beta\,\big]\hat{v}_\beta + \hat{v}_\beta\big[\,\hat{v}_\alpha,\hat{v}_\beta\,\big]\Big)\\
&= \frac{i\hbar q}{\mu}\sum_{\beta\gamma}\epsilon_{\alpha\beta\gamma}\hat{v}_\beta B_\gamma = \frac{i\hbar q}{\mu}\big(\hat{\vec{v}}\times\vec{B}\,\big)_\alpha.
\end{aligned}$$

Considering $\vec{B} = B\vec{e}_z$, we have

$$\begin{aligned}
&\big[\,\hat{v}_x,\hat{H}\,\big] = \frac{i\hbar qB}{\mu}\hat{v}_y,\\
&\big[\,\hat{v}_y,\hat{H}\,\big] = -\frac{i\hbar qB}{\mu}\hat{v}_x,\\
&\big[\,\hat{v}_z,\hat{H}\,\big] = 0.
\end{aligned}$$

Noting that $\hat{v}_x$, $\hat{v}_y$, and $\hat{v}_z$ do not depend on time explicitly, we have from the Heisenberg equation of motion

$$\begin{aligned}
&\frac{d\hat{v}_x}{dt} = \frac{1}{i\hbar}\big[\,\hat{v}_x,\hat{H}\,\big] = \frac{qB}{\mu}\hat{v}_y,\\
&\frac{d\hat{v}_y}{dt} = \frac{1}{i\hbar}\big[\,\hat{v}_y,\hat{H}\,\big] = -\frac{qB}{\mu}\hat{v}_x,\\
&\frac{d\hat{v}_z}{dt} = \frac{1}{i\hbar}\big[\,\hat{v}_z,\hat{H}\,\big] = 0.
\end{aligned}$$

For $\hat{v}_z$, we have

$$\hat{v}_z(t) = \hat{v}_z(0).$$

For the convenience of solving the equations of motion of $\hat{v}_x$ and $\hat{v}_y$, we differentiate them with respect to t and obtain

$$\frac{d^2\hat{v}_x}{dt^2} = \frac{qB}{\mu}\frac{d\hat{v}_y}{dt} = -\left(\frac{qB}{\mu}\right)^2 \hat{v}_x = -\omega^2\hat{v}_x,$$
$$\frac{d^2\hat{v}_y}{dt^2} = -\frac{qB}{\mu}\frac{d\hat{v}_x}{dt} = -\left(\frac{qB}{\mu}\right)^2 \hat{v}_y = -\omega^2\hat{v}_y,$$

where ω is the cyclotron frequency given by

$$\omega = \frac{qB}{\mu}.$$

The general solutions to the above equations are given by

$$\hat{v}_x(t) = \hat{A}\cos(\omega t) + \hat{B}\sin(\omega t),$$
$$\hat{v}_y(t) = \hat{C}\cos(\omega t) + \hat{D}\sin(\omega t).$$

Their derivatives with respect to t are given by

$$\frac{d\hat{v}_x(t)}{dt} = -\omega\hat{A}\sin(\omega t) + \omega\hat{B}\cos(\omega t) = \omega\hat{v}_y(t) = \omega\hat{C}\cos(\omega t) + \omega\hat{D}\sin(\omega t),$$
$$\frac{d\hat{v}_y(t)}{dt} = -\omega\hat{C}\sin(\omega t) + \omega\hat{D}\cos(\omega t) = -\omega\hat{v}_x(t) = -\omega\hat{A}\cos(\omega t) - \omega\hat{B}\sin(\omega t).$$

Setting $t = 0$ in the above four equations, we obtain

$$\hat{v}_x(0) = \hat{A},$$
$$\hat{v}_y(0) = \hat{C},$$
$$\hat{B} = \hat{C} = \hat{v}_y(0),$$
$$\hat{D} = -\hat{A} = -\hat{v}_x(0).$$

We finally have

$$\hat{v}_x(t) = \hat{v}_x(0)\cos(\omega t) + \hat{v}_y(0)\sin(\omega t),$$
$$\hat{v}_y(t) = \hat{v}_y(0)\cos(\omega t) - \hat{v}_x(0)\sin(\omega t).$$

2. (15 points) The Hamiltonian operator of a two-level quantum system is given by $\hat{H} = -\frac{1}{2}\hbar\omega\big(|0\rangle\langle 0| - |1\rangle\langle 1|\big)$, where $|0\rangle$ and $|1\rangle$ are the orthonormal eigenkets of $\hat{H}$ corresponding respectively to the eigenvalues $-\hbar\omega/2$ and $+\hbar\omega/2$. In terms of $|0\rangle$ and $|1\rangle$, the operators $\hat{c}$ and $\hat{c}^\dagger$ are defined respectively by $\hat{c} = |0\rangle\langle 1|$ and $\hat{c}^\dagger = |1\rangle\langle 0|$. Let $\hat{\alpha} = \hat{c} + \hat{c}^\dagger$ and $\hat{\beta} = -i\big(\hat{c} - \hat{c}^\dagger\big)$. At $t = 0$, the system is in the state described by $|\psi(0)\rangle = \frac{1}{\sqrt{2}}\big(|0\rangle + |1\rangle\big)$.

(1) (4 points) Evaluate the anticommutators $\big\{\hat{c}, \hat{c}^\dagger\big\}$, $\big\{\hat{c}, \hat{c}\big\}$, and $\big\{\hat{c}^\dagger, \hat{c}^\dagger\big\}$.

(2) (4 points) Are $\hat{\alpha}$ and $\hat{\beta}$ constants of motion?

(3) (7 points) Write the state vector $|\psi(t)\rangle$ at time $t > 0$. Find the uncertainties $\Delta\alpha$ and $\Delta\beta$ of $\hat{\alpha}$ and $\hat{\beta}$ in $|\psi(t)\rangle$. Is the uncertainty relation satisfied by $\Delta\alpha$ and $\Delta\beta$?

Formulas provided: $\big\{\hat{A}, \hat{B}\big\} = \hat{A}\hat{B} + \hat{B}\hat{A}$, $i\hbar\frac{d}{dt}\hat{A}_H(t) = \big[\hat{A}_H(t), \hat{H}_H\big] + i\hbar\left(\frac{d}{dt}\hat{A}_S(t)\right)_H$,

$$\Delta A = \sqrt{\langle\hat{A}^2\rangle - \langle\hat{A}\rangle^2},\ \Delta A\Delta B \geq \frac{1}{2}\big|\langle[\,\hat{A}, \hat{B}\,]\rangle\big|.$$

(1) (4 points) For $\{\hat{c}, \hat{c}^\dagger\}$, we have

$$\begin{aligned}\{\hat{c}, \hat{c}^\dagger\} &= \hat{c}\hat{c}^\dagger + \hat{c}^\dagger\hat{c} \\ &= |0\rangle\langle 1|1\rangle\langle 0| + |1\rangle\langle 0|0\rangle\langle 1| \\ &= |0\rangle\langle 0| + |1\rangle\langle 1| \\ &= 1,\end{aligned}$$

where the completeness of the set $\{|0\rangle, |1\rangle\}$, $|0\rangle\langle 0| + |1\rangle\langle 1| = 1$, has been used. For $\{\hat{c}, \hat{c}\}$, we have

$$\begin{aligned}\{\hat{c}, \hat{c}\} &= \hat{c}\hat{c} + \hat{c}\hat{c} \\ &= |0\rangle\langle 1|0\rangle\langle 1| + |0\rangle\langle 1|0\rangle\langle 1| \\ &= 0 + 0 \\ &= 0,\end{aligned}$$

where we have made use of $\langle 1|0\rangle = 0$. For $\{\hat{c}^\dagger, \hat{c}^\dagger\}$, we have

$$\begin{aligned}\{\hat{c}^\dagger, \hat{c}^\dagger\} &= \hat{c}^\dagger\hat{c}^\dagger + \hat{c}^\dagger\hat{c}^\dagger \\ &= |1\rangle\langle 0|1\rangle\langle 0| + |1\rangle\langle 0|1\rangle\langle 0| \\ &= 0 + 0 \\ &= 0,\end{aligned}$$

where we have made use of $\langle 0|1\rangle = 0$.

(2) (4 points) To see if $\hat{\alpha}$ and $\hat{\beta}$ are constants of motion, we need only to check if they commute with the Hamiltonian operator $\hat{H}$ since they do not depend explicitly on time. For $[\,\hat{\alpha}, \hat{H}\,]$, we have

$$\begin{aligned}[\,\hat{\alpha}, \hat{H}\,] &= \hat{\alpha}\hat{H} - \hat{H}\hat{\alpha} \\ &= -\frac{1}{2}\hbar\omega\Big[\big(|0\rangle\langle 1| + |1\rangle\langle 0|\big)\big(|0\rangle\langle 0| - |1\rangle\langle 1|\big) - \big(|0\rangle\langle 0| - |1\rangle\langle 1|\big)\big(|0\rangle\langle 1| + |1\rangle\langle 0|\big)\Big] \\ &= -\frac{1}{2}\hbar\omega\Big[|1\rangle\langle 0| - |0\rangle\langle 1| - |0\rangle\langle 1| + |1\rangle\langle 0|\Big] \\ &= -\hbar\omega\big(|1\rangle\langle 0| - |0\rangle\langle 1|\big) = i\hbar\omega\hat{\beta} \neq 0.\end{aligned}$$

For $[\,\hat{\beta}, \hat{H}\,]$, we have

$$\begin{aligned}[\,\hat{\beta}, \hat{H}\,] &= \hat{\beta}\hat{H} - \hat{H}\hat{\beta} \\ &= \frac{i}{2}\hbar\omega\Big[\big(|0\rangle\langle 1| - |1\rangle\langle 0|\big)\big(|0\rangle\langle 0| - |1\rangle\langle 1|\big) - \big(|0\rangle\langle 0| - |1\rangle\langle 1|\big)\big(|0\rangle\langle 1| - |1\rangle\langle 0|\big)\Big] \\ &= \frac{i}{2}\hbar\omega\Big[-|1\rangle\langle 0| - |0\rangle\langle 1| - |0\rangle\langle 1| - |1\rangle\langle 0|\Big] \\ &= -i\hbar\omega\big(|0\rangle\langle 1| + |1\rangle\langle 0|\big) = -i\hbar\omega\hat{\alpha} \neq 0.\end{aligned}$$

Because $[\,\hat{\alpha}, \hat{H}\,] \neq 0$ and $[\,\hat{\beta}, \hat{H}\,] \neq 0$, $\hat{\alpha}$ and $\hat{\beta}$ are not constants of motion.

(3) (7 points) At $t > 0$, the state vector of the system is given by

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}}\big(\,e^{i\omega t/2}\,|0\rangle + e^{-i\omega t/2}\,|1\rangle\big)$$

The average of $\hat{\alpha}$ in $|\psi(t)\rangle$ is given by

$$\begin{aligned}\langle\hat{\alpha}\rangle &= \langle\psi(t)|\hat{\alpha}|\psi(t)\rangle \\ &= \frac{1}{2}\big(e^{-i\omega t/2}\,\langle 0| + e^{i\omega t/2}\,\langle 1|\big)\big(|0\rangle\langle 1| + |1\rangle\langle 0|\big)\big(e^{i\omega t/2}\,|0\rangle + e^{-i\omega t/2}\,|1\rangle\big) \\ &= \frac{1}{2}\big(e^{-i\omega t/2}\,\langle 1| + e^{i\omega t/2}\,\langle 0|\big)\big(e^{i\omega t/2}\,|0\rangle + e^{-i\omega t/2}\,|1\rangle\big) \\ &= \frac{1}{2}\big(e^{-i\omega t} + e^{i\omega t}\big) = \cos(\omega t).\end{aligned}$$

The average of $\hat{\alpha}^2$ in $|\psi(t)\rangle$ is given by

$$\begin{aligned}
\langle\hat{\alpha}^2\rangle &= \langle\psi(t)|\hat{\alpha}^2|\psi(t)\rangle \\
&= \frac{1}{2}\big(e^{-i\omega t/2}\langle 0| + e^{i\omega t/2}\langle 1|\big)\big(|0\rangle\langle 1| + |1\rangle\langle 0|\big)^2\big(e^{i\omega t/2}|0\rangle + e^{-i\omega t/2}|1\rangle\big) \\
&= \frac{1}{2}\big(e^{-i\omega t/2}\langle 1| + e^{i\omega t/2}\langle 0|\big)\big(e^{i\omega t/2}|1\rangle + e^{-i\omega t/2}|0\rangle\big) \\
&= \frac{1}{2}\big(1+1\big) = 1.
\end{aligned}$$

We thus have

$$\Delta\alpha = \sqrt{\langle\hat{\alpha}^2\rangle - \langle\hat{\alpha}\rangle^2} = \sqrt{1-\cos^2(\omega t)} = |\sin(\omega t)|.$$

The average of $\hat{\beta}$ in $|\psi(t)\rangle$ is given by

$$\begin{aligned}
\langle\hat{\beta}\rangle &= \langle\psi(t)|\hat{\beta}|\psi(t)\rangle \\
&= -\frac{i}{2}\big(e^{-i\omega t/2}\langle 0| + e^{i\omega t/2}\langle 1|\big)\big(|0\rangle\langle 1| - |1\rangle\langle 0|\big)\big(e^{i\omega t/2}|0\rangle + e^{-i\omega t/2}|1\rangle\big) \\
&= -\frac{i}{2}\big(e^{-i\omega t/2}\langle 1| - e^{i\omega t/2}\langle 0|\big)\big(e^{i\omega t/2}|0\rangle + e^{-i\omega t/2}|1\rangle\big) \\
&= -\frac{i}{2}\big(e^{-i\omega t} - e^{i\omega t}\big) = -\sin(\omega t).
\end{aligned}$$

The average of $\hat{\beta}^2$ in $|\psi(t)\rangle$ is given by

$$\begin{aligned}
\langle\hat{\beta}^2\rangle &= \langle\psi(t)|\hat{\beta}^2|\psi(t)\rangle \\
&= -\frac{1}{2}\big(e^{-i\omega t/2}\langle 0| + e^{i\omega t/2}\langle 1|\big)\big(|0\rangle\langle 1| - |1\rangle\langle 0|\big)^2\big(e^{i\omega t/2}|0\rangle + e^{-i\omega t/2}|1\rangle\big) \\
&= -\frac{1}{2}\big(e^{-i\omega t/2}\langle 1| - e^{i\omega t/2}\langle 0|\big)\big(-e^{i\omega t/2}|1\rangle + e^{-i\omega t/2}|0\rangle\big) \\
&= -\frac{1}{2}\big(-1-1\big) = 1.
\end{aligned}$$

We thus have

$$\Delta\beta = \sqrt{\langle\hat{\beta}^2\rangle - \langle\hat{\beta}\rangle^2} = \sqrt{1-\sin^2(\omega t)} = |\cos(\omega t)|.$$

The vale of the product $\Delta\alpha\Delta\beta$ is given by

$$\Delta\alpha\Delta\beta = |\sin(\omega t)| \times |\cos(\omega t)| = |\sin(\omega t)\cos(\omega t)| = \frac{1}{2}|\sin(2\omega t)|.$$

To see how the uncertainty relation is satisfied, we need to find the value of $\langle[\hat{\alpha},\hat{\beta}]\rangle$. For $[\hat{\alpha},\hat{\beta}]$, we have

$$\begin{aligned}
[\hat{\alpha},\hat{\beta}] &= \hat{\alpha}\hat{\beta} - \hat{\beta}\hat{\alpha} \\
&= -i\big(|0\rangle\langle 1| + |1\rangle\langle 0|\big)\big(|0\rangle\langle 1| - |1\rangle\langle 0|\big) + i\big(|0\rangle\langle 1| - |1\rangle\langle 0|\big)\big(|0\rangle\langle 1| + |1\rangle\langle 0|\big) \\
&= -i\big(|1\rangle\langle 1| - |0\rangle\langle 0|\big) + i\big(|0\rangle\langle 0| - |1\rangle\langle 1|\big) \\
&= 2i\big(|0\rangle\langle 0| - |1\rangle\langle 1|\big).
\end{aligned}$$

The value of $\langle[\hat{\alpha},\hat{\beta}]\rangle$ is then given by

$$\begin{aligned}
\langle[\hat{\alpha},\hat{\beta}]\rangle &= i\big(e^{-i\omega t/2}\langle 0| + e^{i\omega t/2}\langle 1|\big)\big(|0\rangle\langle 0| - |1\rangle\langle 1|\big)\big(e^{i\omega t/2}|0\rangle + e^{-i\omega t/2}|1\rangle\big) \\
&= i\big(e^{-i\omega t/2}\langle 0| - e^{i\omega t/2}\langle 1|\big)\big(e^{i\omega t/2}|0\rangle + e^{-i\omega t/2}|1\rangle\big) \\
&= i\big(1-1\big) = 0.
\end{aligned}$$

From the uncertainty relation,

$$\Delta\alpha\Delta\beta \geq \frac{1}{2}\big|\langle[\,\hat{\alpha},\hat{\beta}\,]\rangle\big|,$$

we have

$$\Delta\alpha\Delta\beta \geq 0$$

which indicates that the obtained result $\Delta\alpha\Delta\beta = \frac{1}{2}|\sin(2\omega t)|$ satisfies the uncertainty relation.

3. (15 points) The two eigenkets of the unperturbed Hamiltonian operator of a two-state system are denoted by $|1\rangle$ and $|2\rangle$. With the perturbation included the Hermitian representation matrix $H = \begin{pmatrix} H_{11} & H_{12} \\ H_{12}^* & H_{22} \end{pmatrix}$ of the total Hamiltonian operator $\hat{H}$ of the system in the orthonormal basis $\{|1\rangle, |2\rangle\}$ is reexpressed as $H = \big(\epsilon I + \vec{\eta}\cdot\vec{\sigma}\big)/2$, where ϵ is a real number, I is the 2×2 unit matrix, the three-components of the constant vector $\vec{\eta}$ are all real numbers, and $\vec{\sigma} = \sigma_x\vec{e}_x + \sigma_y\vec{e}_y + \sigma_z\vec{e}_z$ with σ_x, σ_y, and σ_z Pauli matrices. Let $|\psi(t)\rangle$ be the normalized state vector of the system at time t. The density operator is given by $\hat{\rho}(t) = |\psi(t)\rangle\langle\psi(t)|$.

(1) (3 points) Express the eigenvalues E_1 and E_2 of $\hat{H}$ and the parameters ϵ, η_x, η_y, and η_z in terms of H_{11}, H_{12}, H_{12}^*, and H_{22}. Let $\hbar\omega = |E_1 - E_2|$. What is the value of η^2?

(2) (3 points) Express the density matrix $\rho(t)$ in the basis $\{|1\rangle, |2\rangle\}$ in terms of $\vec{\sigma}$ in the form of const$+\vec{\Psi}(t)\cdot\vec{\sigma}/2$. What are the expressions of the components of $\vec{\Psi}(t)$ in terms of the components of $|\psi(t)\rangle$ in the basis $\{|1\rangle, |2\rangle\}$? What is the value of $[\vec{\Psi}(t)]^2$? Express $\vec{\Psi}(0)$ as an average in $|\psi(0)\rangle$.

(3) (3 points) Find the equation of motion for $\vec{\Psi}(t)$ from the equation of motion of $\rho(t)$.

(4) (4 points) Solve for $\vec{\Psi}(t)$ from the equation of motion of $\vec{\Psi}(t)$ under the condition that $\vec{\Psi}(t=0) = \vec{\Psi}(0)$.

(5) (2 points) What is the probability of finding the system to be in $|\psi(0)\rangle$ at time $t > 0$?

Formulas provided: $\dfrac{d\hat{\rho}}{dt} = \dfrac{1}{i\hbar}\big[\,\hat{H},\hat{\rho}\,\big]$, $\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$, $\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, $\vec{a}\times(\vec{b}\times\vec{c}) = (\vec{a}\cdot\vec{c})\vec{b} - (\vec{a}\cdot\vec{b})\vec{c}$,

$$M = a_0 I + \vec{a}\cdot\vec{\sigma},\ a_0 = \frac{1}{2}\,\mathrm{Tr}(M),\ a_\alpha = \frac{1}{2}\,\mathrm{Tr}(M\sigma_\alpha),\ \big[\,\sigma_\alpha,\sigma_\beta\,\big] = 2i\sum_\gamma \varepsilon_{\alpha\beta\gamma}\sigma_\gamma.$$

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(1) (3 points) Let E be an eigenvalue of $\hat{H}$. The secular equation is then given by

$$\begin{vmatrix} H_{11} - E & H_{12} \\ H_{12}^* & H_{22} - E \end{vmatrix} = 0.$$

That is,

$$E^2 - (H_{11} + H_{22})E + H_{11}H_{22} - |H_{12}|^2 = 0$$

from which we obtain the following two eigenvalues of $\hat{H}$

$$E_{1,2} = \frac{1}{2}\big[\,H_{11} + H_{22} \pm \sqrt{(H_{11} - H_{22})^2 + 4|H_{12}|^2}\,\big].$$

For ϵ, we have

$$\epsilon = \mathrm{Tr}\,H = H_{11} + H_{22}.$$

For η_x, we have

$$\eta_x = \mathrm{Tr}(H\sigma_x) = \mathrm{Tr}\left[\begin{pmatrix} H_{11} & H_{12} \\ H_{12}^* & H_{22} \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\right] = H_{12} + H_{12}^*.$$

For η_y, we have

$$\eta_y = \mathrm{Tr}(H\sigma_y) = \mathrm{Tr}\Big[\begin{pmatrix} H_{11} & H_{12} \\ H_{12}^* & H_{22} \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \Big] = i\big(H_{12} - H_{12}^*\big).$$

For η_z, we have

$$\eta_z = \mathrm{Tr}(H\sigma_z) = \mathrm{Tr}\Big[\begin{pmatrix} H_{11} & H_{12} \\ H_{12}^* & H_{22} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \Big] = H_{11} - H_{22}.$$

For η^2, we have

$$\begin{aligned} \eta^2 &= \eta_x^2 + \eta_y^2 + \eta_z^2 \\ &= \big(H_{12} + H_{12}^*\big)^2 - \big(H_{12} - H_{12}^*\big)^2 + \big(H_{11} - H_{22}\big)^2 \\ &= \big(H_{11} - H_{22}\big)^2 + 4|H_{12}|^2 \\ &= \big(E_1 - E_2\big)^2 \\ &= (\hbar\omega)^2. \end{aligned}$$

(2) (3 points) In the orthonormal basis $\{|1\rangle, |2\rangle\}$, the matrix elements of $|\psi(t)\rangle$ are given by

$$\psi_1(t) = \langle 1|\psi(t)\rangle,\ \psi_2(t) = \langle 2|\psi(t)\rangle\,.$$

The matrix elements of $\hat{\rho}(t)$ in the orthonormal basis $\{|1\rangle, |2\rangle\}$ are given by

$$\begin{aligned} \langle 1|\hat{\rho}(t)|1\rangle &= \langle 1|\psi(t)\rangle\langle\psi(t)|1\rangle = \psi_1(t)\psi_1^*(t) = |\psi_1(t)|^2, \\ \langle 1|\hat{\rho}(t)|2\rangle &= \langle 1|\psi(t)\rangle\langle\psi(t)|2\rangle = \psi_1(t)\psi_2^*(t) = \psi_2^*(t)\psi_1(t), \\ \langle 2|\hat{\rho}(t)|1\rangle &= \langle 2|\psi(t)\rangle\langle\psi(t)|1\rangle = \psi_2(t)\psi_1^*(t) = \psi_1^*(t)\psi_2(t), \\ \langle 2|\hat{\rho}(t)|2\rangle &= \langle 2|\psi(t)\rangle\langle\psi(t)|2\rangle = \psi_2(t)\psi_2^*(t) = |\psi_2(t)|^2. \end{aligned}$$

Thus, the density matrix $\rho(t)$ in the orthonormal basis $\{|1\rangle, |2\rangle\}$ is given by

$$\rho(t) = \begin{pmatrix} |\psi_1(t)|^2 & \psi_2^*(t)\psi_1(t) \\ \psi_1^*(t)\psi_2(t) & |\psi_2(t)|^2 \end{pmatrix}.$$

Let

$$\rho(t) = a_0(t)I + \vec{a}_1(t)\cdot\vec{\sigma}.$$

For $a_0(t)$, we have

$$a_0(t) = \frac{1}{2}\,\mathrm{Tr}\,\rho(t) = \frac{1}{2}\big[\,|\psi_1(t)|^2 + |\psi_2(t)|^2\,\big] = \frac{1}{2}.$$

For $\vec{a}_1(t)$, we have

$$\begin{aligned} a_{1x}(t) &= \frac{1}{2}\,\mathrm{Tr}\big[\,\rho(t)\sigma_x\,\big] = \frac{1}{2}\,\mathrm{Tr}\Big[\begin{pmatrix} |\psi_1(t)|^2 & \psi_2^*(t)\psi_1(t) \\ \psi_1^*(t)\psi_2(t) & |\psi_2(t)|^2 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \Big] = \frac{1}{2}\big[\,\psi_2^*(t)\psi_1(t) + \psi_1^*(t)\psi_2(t)\,\big], \\ a_{1y}(t) &= \frac{1}{2}\,\mathrm{Tr}\big[\,\rho(t)\sigma_y\,\big] = \frac{1}{2}\,\mathrm{Tr}\Big[\begin{pmatrix} |\psi_1(t)|^2 & \psi_2^*(t)\psi_1(t) \\ \psi_1^*(t)\psi_2(t) & |\psi_2(t)|^2 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \Big] = \frac{i}{2}\big[\,\psi_2^*(t)\psi_1(t) - \psi_1^*(t)\psi_2(t)\,\big], \\ a_{1z}(t) &= \frac{1}{2}\,\mathrm{Tr}\big[\,\rho(t)\sigma_z\,\big] = \frac{1}{2}\,\mathrm{Tr}\Big[\begin{pmatrix} |\psi_1(t)|^2 & \psi_2^*(t)\psi_1(t) \\ \psi_1^*(t)\psi_2(t) & |\psi_2(t)|^2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \Big] = \frac{1}{2}\big[\,|\psi_1(t)|^2 - |\psi_2(t)|^2\,\big]. \end{aligned}$$

We thus have

$$\begin{aligned} \rho(t) &= \frac{1}{2}I + \frac{1}{2}\big[\,\psi_2^*(t)\psi_1(t) + \psi_1^*(t)\psi_2(t)\,\big]\sigma_x + \frac{i}{2}\big[\,\psi_2^*(t)\psi_1(t) - \psi_1^*(t)\psi_2(t)\,\big]\sigma_y + \frac{1}{2}\big[\,|\psi_1(t)|^2 - |\psi_2(t)|^2\,\big]\sigma_z \\ &= \frac{1}{2}\big[\,I + \vec{\Psi}(t)\cdot\vec{\sigma}\,\big] \end{aligned}$$

with

$$\Psi_x(t) = \psi_2^*(t)\psi_1(t) + \psi_1^*(t)\psi_2(t),$$
$$\Psi_y(t) = i\big[\,\psi_2^*(t)\psi_1(t) - \psi_1^*(t)\psi_2(t)\,\big],$$
$$\Psi_z(t) = |\psi_1(t)|^2 - |\psi_2(t)|^2.$$

For $[\vec{\Psi}(t)]^2$, we have

$$\begin{aligned}
[\vec{\Psi}(t)]^2 &= \big[\,\psi_2^*(t)\psi_1(t) + \psi_1^*(t)\psi_2(t)\,\big]^2 - \big[\,\psi_2^*(t)\psi_1(t) - \psi_1^*(t)\psi_2(t)\,\big]^2 + \big[\,|\psi_1(t)|^2 - |\psi_2(t)|^2\,\big]^2 \\
&= |\psi_1(t)|^4 + |\psi_2(t)|^4 + 2|\psi_1(t)|^2|\psi_2(t)|^2 \\
&= \big[\,|\psi_1(t)|^2 + |\psi_2(t)|^2\,\big]^2 \\
&= 1.
\end{aligned}$$

From the above results, we see that

$$\vec{\Psi}(t) = \mathrm{Tr}\big[\,\rho(t)\vec{\sigma}\,\big] = \langle\psi(t)|\hat{\vec{\sigma}}|\psi(t)\rangle\,.$$

For $\vec{\Psi}(0)$, we have

$$\vec{\Psi}(0) = \mathrm{Tr}\big[\,\rho(0)\vec{\sigma}\,\big] = \langle\psi(0)|\hat{\vec{\sigma}}|\psi(0)\rangle\,.$$

(3) (3 points) The equation of motion of $\rho(t)$ reads

$$\frac{d\rho(t)}{dt} = \frac{1}{i\hbar}\big[\,H, \rho(t)\,\big].$$

For the left hand side, we have

$$\frac{d\rho(t)}{dt} = \frac{1}{2}\frac{d\vec{\Psi}(t)}{dt}\cdot\vec{\sigma}.$$

For the right hand side, we have

$$\begin{aligned}
\frac{1}{i\hbar}\big[\,H, \rho(t)\,\big] &= \frac{1}{4i\hbar}\big[\,I + \vec{\eta}\cdot\vec{\sigma}, I + \vec{\Psi}(t)\cdot\vec{\sigma}\big] \\
&= \frac{1}{4i\hbar}\sum_{\alpha\beta}\eta_\alpha\Psi_\beta(t)\big[\,\sigma_\alpha, \sigma_\beta\,\big] \\
&= \frac{1}{2\hbar}\sum_{\alpha\beta\gamma}\varepsilon_{\alpha\beta\gamma}\eta_\alpha\Psi_\beta(t)\sigma_\gamma \\
&= \frac{1}{2\hbar}\big[\,\vec{\eta}\times\vec{\Psi}(t)\,\big]\cdot\vec{\sigma}.
\end{aligned}$$

We then have

$$\frac{1}{2}\frac{d\vec{\Psi}(t)}{dt}\cdot\vec{\sigma} = \frac{1}{2\hbar}\big[\,\vec{\eta}\times\vec{\Psi}(t)\,\big]\cdot\vec{\sigma}$$

from which we obtain

$$\frac{d\vec{\Psi}(t)}{dt} = \frac{1}{\hbar}\,\vec{\eta}\times\vec{\Psi}(t).$$

(4) (4 points) Considering $\big[\,\vec{\eta}\times\vec{\Psi}(t)\,\big]\cdot\vec{\eta} = 0$, we see that the component of $\vec{\Psi}(t)$ in the direction of $\vec{\eta}$ does not vary with time, $\Psi_\parallel(t) = \Psi_\parallel(0)$. For the component of $\vec{\Psi}(t)$ perpendicular to $\vec{\eta}$, $\vec{\Psi}_\perp(t)$, we have

$$\frac{d\vec{\Psi}_\perp(t)}{dt} = \frac{1}{\hbar}\,\vec{\eta}\times\vec{\Psi}_\perp(t).$$

For the purpose of solving the above equation, we differentiate its two sides with respect to t and obtain

$$\begin{aligned}\frac{d^2\vec{\Psi}_\perp(t)}{dt^2} &= \frac{1}{\hbar}\,\vec{\eta}\times\frac{d\vec{\Psi}_\perp(t)}{dt}\\ &= \frac{1}{\hbar^2}\vec{\eta}\times\big[\,\vec{\eta}\times\vec{\Psi}_\perp(t)\,\big]\\ &= -\frac{\eta^2}{\hbar^2}\vec{\Psi}_\perp(t)\\ &= -\omega^2\vec{\Psi}_\perp(t).\end{aligned}$$

The general solution to the above equation is given by

$$\vec{\Psi}_\perp(t) = \vec{A}\cos(\omega t) + \vec{B}\sin(\omega t).$$

From $\vec{\Psi}_\perp(t=0) = \vec{\Psi}_\perp(0)$, we have

$$\vec{A} = \vec{\Psi}_\perp(0).$$

From

$$\frac{d\vec{\Psi}_\perp(t)}{dt} = \frac{1}{\hbar}\,\vec{\eta}\times\vec{\Psi}_\perp(t),$$

we have

$$-\omega\vec{\Psi}_\perp(0)\sin(\omega t) + \omega\vec{B}\cos(\omega t) = \frac{1}{\hbar}\,\vec{\eta}\times\big[\,\vec{\Psi}_\perp(0)\cos(\omega t) + \vec{B}\sin(\omega t)\,\big]$$

from which it follows that

$$\vec{B} = \frac{1}{\hbar\omega}\vec{\eta}\times\vec{\Psi}_\perp(0).$$

We then have

$$\vec{\Psi}_\perp(t) = \vec{\Psi}_\perp(0)\cos(\omega t) + \frac{1}{\hbar\omega}\vec{\eta}\times\vec{\Psi}_\perp(0)\sin(\omega t).$$

For $\vec{\Psi}(t)$, we have

$$\begin{aligned}\vec{\Psi}(t) &= \vec{\Psi}_\parallel(t) + \vec{\Psi}_\perp(t) = \vec{\Psi}_\parallel(0) + \vec{\Psi}_\perp(t)\\ &= \frac{1}{(\hbar\omega)^2}\big[\,\vec{\Psi}(0)\cdot\hat{\vec{\eta}}\,\big]\vec{\eta}\,\big[\,1-\cos(\omega t)\,\big] + \vec{\Psi}(0)\cos(\omega t) + \frac{1}{\hbar\omega}\vec{\eta}\times\vec{\Psi}(0)\sin(\omega t).\end{aligned}$$

Using the unit vector $\hat{\vec{\eta}} = \vec{\eta}/\eta = \vec{\eta}/\hbar\omega$ along the direction of $\vec{\eta}$, we can rewrite $\vec{\Psi}(t)$ as

$$\vec{\Psi}(t) = \big[\,\vec{\Psi}(0)\cdot\hat{\vec{\eta}}\,\big]\hat{\vec{\eta}}\,\big[\,1-\cos(\omega t)\,\big] + \vec{\Psi}(0)\cos(\omega t) + \hat{\vec{\eta}}\times\vec{\Psi}(0)\sin(\omega t).$$

(5) (2 points) The probability of finding the system to be in $|\psi(0)\rangle$ at time $t>0$ is given by

$$\begin{aligned}P_0(t) &= \big|\langle\psi(0)|\psi(t)\rangle\big|^2 = \langle\psi(0)|\psi(t)\rangle\langle\psi(t)|\psi(0)\rangle\\ &= \langle\psi(0)|\hat{\rho}(t)|\psi(0)\rangle = \frac{1}{2} + \frac{1}{2}\,\langle\psi(0)|\vec{\Psi}(t)\cdot\hat{\vec{\sigma}}|\psi(0)\rangle\\ &= \frac{1}{2} + \frac{1}{2}\,\langle\psi(0)|\hat{\vec{\sigma}}|\psi(0)\rangle\cdot\vec{\Psi}(t) = \frac{1}{2} + \frac{1}{2}\vec{\Psi}(0)\cdot\vec{\Psi}(t)\\ &= \frac{1}{2} + \frac{1}{2}\vec{\Psi}(0)\cdot\Big\{\big[\,\vec{\Psi}(0)\cdot\hat{\vec{\eta}}\,\big]\hat{\vec{\eta}}\,\big[\,1-\cos(\omega t)\,\big] + \vec{\Psi}(0)\cos(\omega t) + \hat{\vec{\eta}}\times\vec{\Psi}(0)\sin(\omega t)\Big\}\\ &= \frac{1}{2} + \frac{1}{2}\big[\,\vec{\Psi}(0)\cdot\hat{\vec{\eta}}\,\big]^2\big[\,1-\cos(\omega t)\,\big] + \frac{1}{2}\vec{\Psi}^2(0)\cos(\omega t)\\ &= \frac{1}{2} + \frac{1}{2}\big[\,\vec{\Psi}(0)\cdot\hat{\vec{\eta}}\,\big]^2\big[\,1-\cos(\omega t)\,\big] + \frac{1}{2}\cos(\omega t)\\ &= \cos^2(\omega t/2) + \big[\,\vec{\Psi}(0)\cdot\hat{\vec{\eta}}\,\big]^2\sin^2(\omega t/2).\end{aligned}$$

4. (15 points) The Hamiltonian operator of a forced one-dimensional harmonic oscillator of mass m and angular frequency ω is given by $\hat{H} = \frac{1}{2m}\hat{p}^2 + \frac{1}{2}m\omega^2\hat{x}^2 - g\left(\hat{x} + \frac{1}{m\omega}\hat{p}\right)$, where g is a real number.

(1) (5 points) Express $\hat{H}$ in terms of the particle annihilation and creation operators, $\hat{a}$ and $\hat{a}^\dagger$.

(2) (2 points) Write the Hamiltonian operator, $\hat{H}_H(t)$, of the oscillator in the Heisenberg picture in terms of the particle annihilation and creation operators in the Heisenberg picture, $\hat{a}_H(t)$ and $\hat{a}_H^\dagger(t)$.

(3) (3 points) Find the equation of motion for $\hat{a}_H(t)$ in the Heisenberg picture.

(4) (5 points) Solve the equation of motion of $\hat{a}_H(t)$ under the initial condition that $\hat{a}_H(t=0) = \hat{a}$.

Formulas provided: $\hat{a} = \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} + \frac{i}{m\omega}\hat{p}\right)$, $\hat{a}^\dagger = \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} - \frac{i}{m\omega}\hat{p}\right)$, $\left[\,\hat{a}, \hat{a}^\dagger\,\right] = 1$,

$$\hat{A}_H(t) = \hat{U}^\dagger(t,t_0)\hat{A}_S(t)\hat{U}(t,t_0),\ i\hbar\frac{d}{dt}\hat{A}_H(t) = \left[\hat{A}_H(t), \hat{H}_H\right] + i\hbar\left(\frac{d}{dt}\hat{A}_S(t)\right)_H.$$

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(1) (5 points) Solving for $\hat{x}$ and $\hat{p}$ from the given definitions of $\hat{a}$ and $\hat{a}^\dagger$, we obtain

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}\left(\hat{a} + \hat{a}^\dagger\right),$$
$$\hat{p} = -i\sqrt{\frac{m\hbar\omega}{2}}\left(\hat{a} - \hat{a}^\dagger\right).$$

We first express the factors $\hat{p}^2$, $\hat{x}^2$, and $\hat{x} + \frac{1}{m\omega}\hat{p}$ in $\hat{H}$ in terms of $\hat{a}$ and $\hat{a}^\dagger$. For $\hat{p}^2$, we have

$$\begin{aligned}\hat{p}^2 &= -\frac{m\hbar\omega}{2}\left(\hat{a} - \hat{a}^\dagger\right)^2 \\ &= -\frac{m\hbar\omega}{2}\left(\hat{a}^2 + \hat{a}^{\dagger 2} - \hat{a}\hat{a}^\dagger - \hat{a}^\dagger\hat{a}\right) \\ &= -\frac{m\hbar\omega}{2}\left(\hat{a}^2 + \hat{a}^{\dagger 2} - 2\hat{a}^\dagger\hat{a} - 1\right).\end{aligned}$$

For $\hat{x}^2$, we have

$$\begin{aligned}\hat{x}^2 &= \frac{\hbar}{2m\omega}\left(\hat{a} + \hat{a}^\dagger\right)^2 \\ &= \frac{\hbar}{2m\omega}\left(\hat{a}^2 + \hat{a}^{\dagger 2} + \hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a}\right) \\ &= \frac{\hbar}{2m\omega}\left(\hat{a}^2 + \hat{a}^{\dagger 2} + 2\hat{a}^\dagger\hat{a} + 1\right).\end{aligned}$$

For $\hat{x} + \frac{1}{m\omega}\hat{p}$, we have

$$\begin{aligned}\hat{x} + \frac{1}{m\omega}\hat{p} &= \sqrt{\frac{\hbar}{2m\omega}}\left(\hat{a} + \hat{a}^\dagger\right) - i\sqrt{\frac{\hbar}{2m\omega}}\left(\hat{a} - \hat{a}^\dagger\right) \\ &= \sqrt{\frac{\hbar}{2m\omega}}\left[\left(1-i\right)\hat{a} + \left(1+i\right)\hat{a}^\dagger\right].\end{aligned}$$

Inserting the above results into $\hat{H}$, we obtain

$$\begin{aligned}\hat{H} = &-\frac{1}{2m} \times \frac{m\hbar\omega}{2}\left(\hat{a}^2 + \hat{a}^{\dagger 2} - 2\hat{a}^\dagger\hat{a} - 1\right) + \frac{1}{2}m\omega^2 \times \frac{\hbar}{2m\omega}\left(\hat{a}^2 + \hat{a}^{\dagger 2} + 2\hat{a}^\dagger\hat{a} + 1\right) \\ &- g\sqrt{\frac{\hbar}{2m\omega}}\left[\left(1-i\right)\hat{a} + \left(1+i\right)\hat{a}^\dagger\right] \\ = &\ \hbar\omega\left(\hat{a}^\dagger\hat{a} + 1/2\right) + \gamma^*\hat{a} + \gamma\hat{a}^\dagger,\end{aligned}$$

where

$$\gamma = -g\sqrt{\frac{\hbar}{2m\omega}}(1+i).$$

(2) (2 points) Setting $t_0 = 0$, we have $\hat{U}(t,0) = e^{-i\hat{H}t/\hbar}$. To obtain the Hamiltonian operator in the Heisenberg picture, we make a unitary transformation to the above-derived Hamiltonian operator in the Schrödinger picture using $\hat{U}(t,0)$. We have

$$\begin{aligned}\hat{U}^\dagger(t,0)\hat{H}\hat{U}(t,0) &= \hbar\omega\big[\,\hat{U}^\dagger(t,0)\hat{a}^\dagger\hat{a}\hat{U}(t,0) + 1/2\,\big] + \gamma^*\hat{U}^\dagger(t,0)\hat{a}\hat{U}(t,0) + \gamma\hat{U}^\dagger(t,0)\hat{a}^\dagger\hat{U}(t,0)\\ &= \hbar\omega\big[\,\hat{U}^\dagger(t,0)\hat{a}^\dagger\hat{U}(t,0)\hat{U}^\dagger(t,0)\hat{a}\hat{U}(t,0) + 1/2\,\big] + \gamma^*\hat{U}^\dagger(t,0)\hat{a}\hat{U}(t,0) + \gamma\hat{U}^\dagger(t,0)\hat{a}^\dagger\hat{U}(t,0),\end{aligned}$$

where we have inserted $\hat{U}(t,0)\hat{U}^\dagger(t,0) = 1$ between $\hat{a}^\dagger$ and $\hat{a}$ in the product $\hat{a}^\dagger\hat{a}$. Making use of

$$\begin{aligned}\hat{H}_H(t) &= \hat{U}^\dagger(t,0)\hat{H}\hat{U}(t,0),\\ \hat{a}^\dagger_H(t) &= \hat{U}^\dagger(t,0)\hat{a}^\dagger\hat{U}(t,0),\\ \hat{a}_H(t) &= \hat{U}^\dagger(t,0)\hat{a}\hat{U}(t,0),\end{aligned}$$

we have the following Hamiltonian operator in the Heisenberg picture

$$\hat{H}_H(t) = \hbar\omega\big[\,\hat{a}^\dagger_H(t)\hat{a}_H(t) + 1/2\,\big] + \gamma^*\hat{a}_H(t) + \gamma\hat{a}^\dagger_H(t).$$

(3) (3 points) Since $\hat{a}$ does not depend on time explicitly in the Schrödinger picture, the Heisenberg equation of motion for $\hat{a}_H(t)$ becomes

$$\frac{d\hat{a}_H(t)}{dt} = \frac{1}{i\hbar}\big[\,\hat{a}_H(t), \hat{H}_H(t)\,\big].$$

Inserting $\hat{H}_H(t) = \hat{U}^\dagger(t,0)\hat{H}\hat{U}(t,0)$ and $\hat{a}_H(t) = \hat{U}^\dagger(t,0)\hat{a}\hat{U}(t,0)$ into the commutator $\big[\,\hat{a}_H(t), \hat{H}_H(t)\,\big]$ yields

$$\begin{aligned}\big[\,\hat{a}_H(t), \hat{H}_H(t)\,\big] &= \big[\,\hat{U}^\dagger(t,0)\hat{a}\hat{U}(t,0), \hat{U}^\dagger(t,0)\hat{H}\hat{U}(t,0)\,\big]\\ &= \hat{U}^\dagger(t,0)\hat{a}\hat{U}(t,0)\hat{U}^\dagger(t,0)\hat{H}\hat{U}(t,0) - \hat{U}^\dagger(t,0)\hat{H}\hat{U}(t,0)\hat{U}^\dagger(t,0)\hat{a}\hat{U}(t,0)\\ &= \hat{U}^\dagger(t,0)\big(\hat{a}\hat{H} - \hat{H}\hat{a}\big)\hat{U}(t,0)\\ &= \hat{U}^\dagger(t,0)\big[\,\hat{a}, \hat{H}\,\big]\hat{U}(t,0),\end{aligned}$$

where we have made use of $\hat{U}(t,0)\hat{U}^\dagger(t,0) = 1$. The commutator $\big[\,\hat{a}, \hat{H}\,\big]$ can be evaluated as follows

$$\begin{aligned}\big[\,\hat{a}, \hat{H}\,\big] &= \big[\,\hat{a}, \hbar\omega\big(\hat{a}^\dagger\hat{a} + 1/2\big) + \gamma^*\hat{a} + \gamma\hat{a}^\dagger\,\big]\\ &= \hbar\omega\big[\,\hat{a}, \hat{a}^\dagger\hat{a}\,\big] + \gamma\\ &= \hbar\omega\Big(\big[\,\hat{a}, \hat{a}^\dagger\,\big]\hat{a} + \hat{a}^\dagger\big[\,\hat{a}, \hat{a}\,\big]\Big) + \gamma\\ &= \hbar\omega\hat{a} + \gamma.\end{aligned}$$

We then have the following equation of motion for $\hat{a}_H(t)$

$$\frac{d\hat{a}_H(t)}{dt} = -i\omega\hat{a}_H(t) - \frac{i\gamma}{\hbar}.$$

(4) (5 points) Considering the presence of the term $-i\omega\hat{a}_H(t)$ on the right hand side of the equation of motion of $\hat{a}_H(t)$, we set $\hat{a}_H(t) = e^{-i\omega t}\hat{\alpha}(t)$ with $\hat{\alpha}(t)$ to be determined. We then have

$$-i\omega e^{-i\omega t}\hat{\alpha} + e^{-i\omega t}\frac{d\hat{\alpha}}{dt} = -i\omega e^{-i\omega t}\hat{\alpha} - \frac{i\gamma}{\hbar}$$

which is simplified to

$$\frac{d\hat{\alpha}(t)}{dt} = -\frac{i\gamma}{\hbar}e^{i\omega t}.$$

Under the condition that $\hat{\alpha}(t=0) = \hat{a}$ from the fact that $\hat{a}_H(t=0) = \hat{a}$, the integration of the above equation yields

$$\begin{aligned}\hat{\alpha}(t) &= \hat{a} - \frac{i\gamma}{\hbar}\int_0^t dt'\, e^{i\omega t'} \\ &= \hat{a} - \frac{\gamma}{\hbar\omega}\big(e^{i\omega t} - 1\big) \\ &= \hat{a} - \frac{2i\gamma}{\hbar\omega} e^{i\omega t/2}\sin(\omega t).\end{aligned}$$

We finally have

$$\begin{aligned}\hat{a}_H(t) &= e^{-i\omega t}\hat{\alpha}(t) \\ &= e^{-i\omega t}\hat{a} - \frac{2i\gamma}{\hbar\omega} e^{-i\omega t/2}\sin(\omega t).\end{aligned}$$

5. (15 points) Two identical one-dimensional harmonic oscillators of mass m and angular frequency ω are coupled together with the interaction Hamiltonian operator given by $\hat{H}_{\rm int} = \lambda m\omega^2 \hat{x}_1\hat{x}_2$, where λ is a real constant with $|\lambda| \ll 1$ and $\hat{x}_1$ and $\hat{x}_2$ are respectively the position operators of the two oscillators. The total Hamiltonian operator of the system is then given by $\hat{H} = \hat{H}_0 + \hat{H}_{\rm int}$ with $\hat{H}_0 = \frac{\hat{p}_1^2}{2m} + \frac{1}{2}m\omega^2\hat{x}_1^2 + \frac{\hat{p}_2^2}{2m} + \frac{1}{2}m\omega^2\hat{x}_2^2$. For the purpose of diagonalizing $\hat{H}$, we introduce operators $\hat{X} = \frac{1}{\sqrt{2}}\big(\hat{x}_1 + \hat{x}_2\big)$, $\hat{P} = \frac{1}{\sqrt{2}}\big(\hat{p}_1 + \hat{p}_2\big)$, $\hat{x} = \frac{1}{\sqrt{2}}\big(\hat{x}_1 - \hat{x}_2\big)$, and $\hat{p} = \frac{1}{\sqrt{2}}\big(\hat{p}_1 - \hat{p}_2\big)$. Obviously, $\big[\,\hat{X},\hat{x}\,\big] = \big[\,\hat{P},\hat{p}\,\big] = 0$

(1) (5 points) Evaluate the commutators $\big[\,\hat{X},\hat{P}\,\big]$, $\big[\,\hat{x},\hat{p}\,\big]$, $\big[\,\hat{X},\hat{p}\,\big]$, and $\big[\,\hat{x},\hat{P}\,\big]$.

(2) (7 points) Express $\hat{H}$ in terms of $\hat{X}$, $\hat{P}$, $\hat{x}$, and $\hat{p}$.

(3) (3 points) Write the energy eigenvalues of $\hat{H}$ by making use of the eigenvalues $(n+1/2)\hbar\omega$ of the Hamiltonian operator of a single isolated one-dimensional harmonic oscillator.

Formula provided: $\big[\,\hat{x}_\alpha,\hat{p}_\beta\,\big] = i\hbar\delta_{\alpha\beta}$.

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(1) (5 points) For the commutator $\big[\,\hat{X},\hat{P}\,\big]$, we have

$$\big[\,\hat{X},\hat{P}\,\big] = \frac{1}{2}\big[\,\hat{x}_1 + \hat{x}_2, \hat{p}_1 + \hat{p}_2\,\big] = \frac{1}{2}\Big\{\big[\,\hat{x}_1,\hat{p}_1\,\big] + \big[\,\hat{x}_2,\hat{p}_2\,\big]\Big\} = \frac{1}{2}\big(i\hbar + i\hbar\big) = i\hbar.$$

For the commutator $\big[\,\hat{x},\hat{p}\,\big]$, we have

$$\big[\,\hat{x},\hat{p}\,\big] = \frac{1}{2}\big[\,\hat{x}_1 - \hat{x}_2, \hat{p}_1 - \hat{p}_2\,\big] = \frac{1}{2}\Big\{\big[\,\hat{x}_1,\hat{p}_1\,\big] + \big[\,\hat{x}_2,\hat{p}_2\,\big]\Big\} = \frac{1}{2}\big(i\hbar + i\hbar\big) = i\hbar.$$

For the commutator $\big[\,\hat{X},\hat{p}\,\big]$, we have

$$\big[\,\hat{X},\hat{p}\,\big] = \frac{1}{2}\big[\,\hat{x}_1 + \hat{x}_2, \hat{p}_1 - \hat{p}_2\,\big] = \frac{1}{2}\Big\{\big[\,\hat{x}_1,\hat{p}_1\,\big] - \big[\,\hat{x}_2,\hat{p}_2\,\big]\Big\} = \frac{1}{2}\big(i\hbar - i\hbar\big) = 0.$$

For the commutator $\big[\,\hat{x},\hat{P}\,\big]$, we have

$$\big[\,\hat{x},\hat{P}\,\big] = \frac{1}{2}\big[\,\hat{x}_1 - \hat{x}_2, \hat{p}_1 + \hat{p}_2\,\big] = \frac{1}{2}\Big\{\big[\,\hat{x}_1,\hat{p}_1\,\big] - \big[\,\hat{x}_2,\hat{p}_2\,\big]\Big\} = \frac{1}{2}\big(i\hbar - i\hbar\big) = 0.$$

(2) (7 points) Expressing the old operators in terms of the new operators, we have

$$\begin{aligned}\hat{x}_1 &= \frac{1}{\sqrt{2}}\big(\hat{X} + \hat{x}\big),\ \hat{x}_2 = \frac{1}{\sqrt{2}}\big(\hat{X} - \hat{x}\big), \\ \hat{p}_1 &= \frac{1}{\sqrt{2}}\big(\hat{P} + \hat{p}\big),\ \hat{p}_2 = \frac{1}{\sqrt{2}}\big(\hat{P} - \hat{p}\big).\end{aligned}$$

Expressing $\hat{H}_{\rm int}$ in terms of $\hat{X}$ and $\hat{x}$, we have

$$\hat{H}_{\rm int} = \frac{1}{2}\lambda m\omega^2\big(\hat{X}+\hat{x}\big)\big(\hat{X}-\hat{x}\big) = \frac{1}{2}\lambda m\omega^2\big(\hat{X}^2-\hat{x}^2\big).$$

Expressing $\hat{H}_0$ in terms of $\hat{X}$, $\hat{P}$, $\hat{x}$, and $\hat{p}$, we have

$$\begin{aligned}\hat{H}_0 &= \frac{1}{4m}\big(\hat{P}+\hat{p}\big)^2 + \frac{1}{4}m\omega^2\big(\hat{X}+\hat{x}\big)^2 + \frac{1}{4m}\big(\hat{P}-\hat{p}\big)^2 + \frac{1}{4}m\omega^2\big(\hat{X}+\hat{x}\big)^2 \\ &= \frac{\hat{P}^2}{2m} + \frac{1}{2}m\omega^2\hat{X}^2 + \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2.\end{aligned}$$

We then have

$$\begin{aligned}\hat{H} &= \frac{\hat{P}^2}{2m} + \frac{1}{2}\big(1+\lambda\big)m\omega^2\hat{X}^2 + \frac{\hat{p}^2}{2m} + \frac{1}{2}(1-\lambda)m\omega^2\hat{x}^2 \\ &= \hat{H}_X + \hat{H}_x,\end{aligned}$$

where

$$\begin{aligned}\hat{H}_X &= \frac{\hat{P}^2}{2m} + \frac{1}{2}m\omega_X^2\hat{X}^2, \\ \hat{H}_x &= \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega_x^2\hat{x}^2\end{aligned}$$

with

$$\begin{aligned}\omega_X &= \omega\sqrt{1+\lambda}, \\ \omega_x &= \omega\sqrt{1-\lambda}.\end{aligned}$$

(3) (3 points) From the commutation relations

$$\begin{aligned}\big[\,\hat{X},\hat{x}\,\big] &= \big[\,\hat{P},\hat{p}\,\big] = 0, \\ \big[\,\hat{X},\hat{p}\,\big] &= \big[\,\hat{x},\hat{P}\,\big] = 0,\end{aligned}$$

We see that

$$\big[\,\hat{H}_X,\hat{H}_x\,\big] = \big[\,\hat{H},\hat{H}_X\,\big] = \big[\,\hat{H},\hat{H}_x\,\big] = 0.$$

Thus, $\hat{H}_X$ and $\hat{H}_x$ can be diagonalized separately. From the eigenvalues $(n+1/2)\hbar\omega$ of the Hamiltonian operator of a single isolated one-dimensional harmonic oscillator, we immediately see that the eigenvalues of $\hat{H}_X$ and $\hat{H}_x$ are given respectively by

$$\begin{aligned}\mathcal{E}_{n_1} &= (n_1+1/2)\hbar\omega_X,\ n_1 = 0,1,2,\cdots, \\ \varepsilon_{n_2} &= (n_2+1/2)\hbar\omega_x,\ n_2 = 0,1,2,\cdots.\end{aligned}$$

The eigenvalues of $\hat{H}$ are then given by

$$E_{n_1n_2} = \mathcal{E}_{n_1} + \varepsilon_{n_2} = (n_1+1/2)\hbar\omega_X + (n_2+1/2)\hbar\omega_x,\ n_1,n_2 = 0,1,2,\cdots$$

6. (15 points) The operators $\hat{a}_1$, $\hat{a}_1^\dagger$ and $\hat{a}_2$, $\hat{a}_2^\dagger$ satisfy the commutation relations $\big[\,\hat{a}_\alpha,\hat{a}_\beta^\dagger\,\big] = \delta_{\alpha\beta}$ and $\big[\,\hat{a}_\alpha,\hat{a}_\beta\,\big] = \big[\,\hat{a}_\alpha^\dagger,\hat{a}_\beta^\dagger\,\big] = 0$ with $\alpha,\beta = 1,2$. In terms of the operators $\hat{a}_1$, $\hat{a}_1^\dagger$ and $\hat{a}_2$, $\hat{a}_2^\dagger$, the following operators are introduced

$$\hat{J}_1 = \frac{\hbar}{2}\big(\hat{a}_2^\dagger\hat{a}_1 + \hat{a}_1^\dagger\hat{a}_2\big),\ \hat{J}_2 = \frac{i\hbar}{2}\big(\hat{a}_2^\dagger\hat{a}_1 - \hat{a}_1^\dagger\hat{a}_2\big),\ \hat{J}_3 = \frac{\hbar}{2}\big(\hat{a}_1^\dagger\hat{a}_1 - \hat{a}_2^\dagger\hat{a}_2\big),\ \hat{j} = \frac{1}{2}\big(\hat{a}_1^\dagger\hat{a}_1 + \hat{a}_2^\dagger\hat{a}_2\big).$$

(1) (6 points) Evaluate the commutators $\big[\,\hat{J}_1,\hat{J}_2\,\big]$, $\big[\,\hat{J}_2,\hat{J}_3\,\big]$, and $\big[\,\hat{J}_3,\hat{J}_1\,\big]$. What do $\hat{J}_1$, $\hat{J}_2$, and $\hat{J}_3$ represent?

(2) (3 points) Express $\hat{\vec{J}}^2 = \hat{J}_1^2 + \hat{J}_2^2 + \hat{J}_3^2$ in terms of $\hat{j}$.

(3) (6 points) Let $|n_1 n_2\rangle$ be the common eigenvectors of $\hat{N}_1 = \hat{a}_1^\dagger \hat{a}_1$ and $\hat{N}_2 = \hat{a}_2^\dagger \hat{a}_2$ with $n_1, n_2 = 0, 1, 2, \cdots$. Evaluate $\hat{J}_3 |n_1 n_2\rangle$, $\hat{\vec{J}}^2 |n_1 n_2\rangle$, and $\big(\hat{J}_1 \pm i\hat{J}_2\big) |n_1 n_2\rangle$. Rewrite the obtained results in the conventional forms through introducing proper quantum numbers.

Formulas provided: $\big[\,\hat{A}, \hat{B}\hat{C}\,\big] = \hat{B}\big[\,\hat{A}, \hat{C}\,\big] + \big[\,\hat{A}, \hat{B}\,\big]\hat{C},\ \big[\,\hat{A}\hat{B}, \hat{C}\,\big] = \hat{A}\big[\,\hat{B}, \hat{C}\,\big] + \big[\,\hat{A}, \hat{C}\,\big]\hat{B},$
$\hat{N}_1 |n_1 n_2\rangle = n_1 |n_1 n_2\rangle,\ \hat{a}_1 |n_1 n_2\rangle = \sqrt{n_1}\, |n_1 - 1, n_2\rangle,\ \hat{a}_1^\dagger |n_1 n_2\rangle = \sqrt{n_1 + 1}\, |n_1 + 1, n_2\rangle,$
$\hat{N}_2 |n_1 n_2\rangle = n_2 |n_1 n_2\rangle,\ \hat{a}_2 |n_1 n_2\rangle = \sqrt{n_2}\, |n_1, n_2 - 1\rangle,\ \hat{a}_2^\dagger |n_1 n_2\rangle = \sqrt{n_2 + 1}\, |n_1, n_2 + 1\rangle.$

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(1) (6 points) For $\big[\,\hat{J}_1, \hat{J}_2\,\big]$, we have

$$\begin{aligned}
\big[\,\hat{J}_1, \hat{J}_2\,\big] &= \frac{i\hbar^2}{4}\big[\,\hat{a}_2^\dagger \hat{a}_1 + \hat{a}_1^\dagger \hat{a}_2, \hat{a}_2^\dagger \hat{a}_1 - \hat{a}_1^\dagger \hat{a}_2\,\big] \\
&= \frac{i\hbar^2}{4}\Big(-\big[\,\hat{a}_2^\dagger \hat{a}_1, \hat{a}_1^\dagger \hat{a}_2\,\big] + \big[\,\hat{a}_1^\dagger \hat{a}_2, \hat{a}_2^\dagger \hat{a}_1\,\big]\Big) \\
&= \frac{i\hbar^2}{4}\Big(-\hat{a}_2^\dagger\big[\,\hat{a}_1, \hat{a}_1^\dagger,\big]\hat{a}_2\ - \hat{a}_1^\dagger\big[\,\hat{a}_2^\dagger, \hat{a}_2\,\big]\hat{a}_1 + \hat{a}_2^\dagger\big[\,\hat{a}_1^\dagger, \hat{a}_1\,\big]\hat{a}_2 + \hat{a}_1^\dagger\big[\,\hat{a}_2, \hat{a}_2^\dagger\big]\hat{a}_1\Big) \\
&= \frac{i\hbar^2}{4}\Big(-\hat{a}_2^\dagger \hat{a}_2 + \hat{a}_1^\dagger \hat{a}_1 - \hat{a}_2^\dagger \hat{a}_2 + \hat{a}_1^\dagger \hat{a}_1\Big) \\
&= \frac{i\hbar^2}{2}\Big(\hat{a}_1^\dagger \hat{a}_1 - \hat{a}_2^\dagger \hat{a}_2\Big) \\
&= i\hbar \hat{J}_3.
\end{aligned}$$

For $\big[\,\hat{J}_2, \hat{J}_3\,\big]$, we have

$$\begin{aligned}
\big[\,\hat{J}_2, \hat{J}_3\,\big] &= \frac{i\hbar^2}{4}\big[\,\hat{a}_2^\dagger \hat{a}_1 - \hat{a}_1^\dagger \hat{a}_2, \hat{a}_1^\dagger \hat{a}_1 - \hat{a}_2^\dagger \hat{a}_2\,\big] \\
&= \frac{i\hbar^2}{4}\Big(\big[\,\hat{a}_2^\dagger \hat{a}_1, \hat{a}_1^\dagger \hat{a}_1\,\big] - \big[\,\hat{a}_2^\dagger \hat{a}_1, \hat{a}_2^\dagger \hat{a}_2\,\big] - \big[\,\hat{a}_1^\dagger \hat{a}_2, \hat{a}_1^\dagger \hat{a}_1\,\big] + \big[\,\hat{a}_1^\dagger \hat{a}_2, \hat{a}_2^\dagger \hat{a}_2\,\big]\Big) \\
&= \frac{i\hbar^2}{4}\Big(\hat{a}_2^\dagger\big[\,\hat{a}_1, \hat{a}_1^\dagger\,\big]\hat{a}_1 - \hat{a}_2^\dagger\big[\,\hat{a}_2^\dagger, \hat{a}_2\,\big]\hat{a}_1 - \hat{a}_1^\dagger\big[\,\hat{a}_1^\dagger, \hat{a}_1\,\big]\hat{a}_2 + \hat{a}_1^\dagger\big[\,\hat{a}_2, \hat{a}_2^\dagger\,\big]\hat{a}_2\Big) \\
&= \frac{i\hbar^2}{4}\Big(\hat{a}_2^\dagger \hat{a}_1 + \hat{a}_2^\dagger \hat{a}_1 + \hat{a}_1^\dagger \hat{a}_2 + \hat{a}_1^\dagger \hat{a}_2\Big) \\
&= \frac{i\hbar^2}{2}\Big(\hat{a}_2^\dagger \hat{a}_1 + \hat{a}_1^\dagger \hat{a}_2\Big) \\
&= i\hbar \hat{J}_1.
\end{aligned}$$

For $\big[\,\hat{J}_3, \hat{J}_1\,\big]$, we have

$$\begin{aligned}
\big[\,\hat{J}_3, \hat{J}_1\,\big] &= \frac{\hbar^2}{4}\big[\,\hat{a}_1^\dagger \hat{a}_1 - \hat{a}_2^\dagger \hat{a}_2, \hat{a}_2^\dagger \hat{a}_1 + \hat{a}_1^\dagger \hat{a}_2\,\big] \\
&= \frac{\hbar^2}{4}\Big(\big[\,\hat{a}_1^\dagger \hat{a}_1, \hat{a}_2^\dagger \hat{a}_1\,\big] + \big[\,\hat{a}_1^\dagger \hat{a}_1, \hat{a}_1^\dagger \hat{a}_2\,\big] - \big[\,\hat{a}_2^\dagger \hat{a}_2, \hat{a}_2^\dagger \hat{a}_1\,\big] - \big[\,\hat{a}_2^\dagger \hat{a}_2, \hat{a}_1^\dagger \hat{a}_2\,\big]\Big) \\
&= \frac{\hbar^2}{4}\Big(\hat{a}_2^\dagger\big[\,\hat{a}_1^\dagger, \hat{a}_1\,\big]\hat{a}_1 + \hat{a}_1^\dagger\big[\,\hat{a}_1, \hat{a}_1^\dagger\,\big]\hat{a}_2 - \hat{a}_2^\dagger\big[\,\hat{a}_2, \hat{a}_2^\dagger\big]\hat{a}_1 - \hat{a}_1^\dagger\big[\,\hat{a}_2^\dagger, \hat{a}_2\,\big]\hat{a}_2\Big) \\
&= \frac{\hbar^2}{4}\Big(-\hat{a}_2^\dagger \hat{a}_1 + \hat{a}_1^\dagger \hat{a}_2 - \hat{a}_2^\dagger \hat{a}_1 + \hat{a}_1^\dagger \hat{a}_2\Big) \\
&= -\frac{\hbar^2}{2}\Big(\hat{a}_2^\dagger \hat{a}_1 - \hat{a}_1^\dagger \hat{a}_2\Big) \\
&= i\hbar \hat{J}_2.
\end{aligned}$$

Because the commutators of $\hat{J}_1$, $\hat{J}_2$, and $\hat{J}_3$ are exactly those of the component operators of an angular momentum, $\hat{J}_1$, $\hat{J}_2$, and $\hat{J}_3$ are the component operators of an angular momentum.

(2) (3 points) For $\hat{\vec{J}}^2$, we have

$$\begin{aligned}
\hat{\vec{J}}^2 &= \hat{J}_1^2 + \hat{J}_2^2 + \hat{J}_3^2 \\
&= \frac{\hbar^2}{4}\big(\hat{a}_2^\dagger\hat{a}_1 + \hat{a}_1^\dagger\hat{a}_2\big)^2 - \frac{\hbar^2}{4}\big(\hat{a}_2^\dagger\hat{a}_1 - \hat{a}_1^\dagger\hat{a}_2\big)^2 + \frac{\hbar^2}{4}\big(\hat{a}_1^\dagger\hat{a}_1 - \hat{a}_2^\dagger\hat{a}_2\big)^2 \\
&= \frac{\hbar^2}{4}\big(2\hat{a}_2^\dagger\hat{a}_1\hat{a}_1^\dagger\hat{a}_2 + 2\hat{a}_1^\dagger\hat{a}_2\hat{a}_2^\dagger\hat{a}_1\big) + \frac{\hbar^2}{4}\big(\hat{a}_1^\dagger\hat{a}_1\hat{a}_1^\dagger\hat{a}_1 + \hat{a}_2^\dagger\hat{a}_2\hat{a}_2^\dagger\hat{a}_2 - 2\hat{a}_1^\dagger\hat{a}_1\hat{a}_2^\dagger\hat{a}_2\big) \\
&= \frac{\hbar^2}{4}\big(2\hat{a}_2^\dagger\hat{a}_2 + 2\hat{a}_1^\dagger\hat{a}_1\hat{a}_2^\dagger\hat{a}_2 + 2\hat{a}_1^\dagger\hat{a}_1 + 2\hat{a}_1^\dagger\hat{a}_1\hat{a}_2^\dagger\hat{a}_2\big) + \frac{\hbar^2}{4}\big(\hat{a}_1^\dagger\hat{a}_1\hat{a}_1^\dagger\hat{a}_1 + \hat{a}_2^\dagger\hat{a}_2\hat{a}_2^\dagger\hat{a}_2 - 2\hat{a}_1^\dagger\hat{a}_1\hat{a}_2^\dagger\hat{a}_2\big) \\
&= \frac{\hbar^2}{4}\big(2\hat{a}_2^\dagger\hat{a}_2 + 2\hat{a}_1^\dagger\hat{a}_1\hat{a}_2^\dagger\hat{a}_2 + 2\hat{a}_1^\dagger\hat{a}_1\big) + \frac{\hbar^2}{4}\big(\hat{a}_1^\dagger\hat{a}_1\hat{a}_1^\dagger\hat{a}_1 + \hat{a}_2^\dagger\hat{a}_2\hat{a}_2^\dagger\hat{a}_2\big) \\
&= \frac{\hbar^2}{4}\big(\hat{a}_1^\dagger\hat{a}_1\hat{a}_1^\dagger\hat{a}_1 + \hat{a}_1^\dagger\hat{a}_1\hat{a}_2^\dagger\hat{a}_2 + 2\hat{a}_1^\dagger\hat{a}_1\big) + \frac{\hbar^2}{4}\big(\hat{a}_1^\dagger\hat{a}_1\hat{a}_2^\dagger\hat{a}_2 + \hat{a}_2^\dagger\hat{a}_2\hat{a}_2^\dagger\hat{a}_2 + 2\hat{a}_2^\dagger\hat{a}_2\big) \\
&= \frac{\hbar^2}{4}\hat{a}_1^\dagger\hat{a}_1\big(\hat{a}_1^\dagger\hat{a}_1 + \hat{a}_2^\dagger\hat{a}_2 + 2\big) + \frac{\hbar^2}{4}\hat{a}_2^\dagger\hat{a}_2\big(\hat{a}_1^\dagger\hat{a}_1 + \hat{a}_2^\dagger\hat{a}_2 + 2\big) \\
&= \frac{\hbar^2}{4}\big(\hat{a}_1^\dagger\hat{a}_1 + \hat{a}_2^\dagger\hat{a}_2\big)\big(\hat{a}_1^\dagger\hat{a}_1 + \hat{a}_2^\dagger\hat{a}_2 + 2\big) \\
&= \hbar^2\hat{j}\big(\hat{j}+1\big).
\end{aligned}$$

(3) (6 points) For $\hat{J}_3\,|n_1n_2\rangle$, we have

$$\begin{aligned}
\hat{J}_3\,|n_1n_2\rangle &= \frac{\hbar}{2}\big(\hat{a}_1^\dagger\hat{a}_1 - \hat{a}_2^\dagger\hat{a}_2\big)\,|n_1n_2\rangle \\
&= \frac{\hbar}{2}\big(n_1 - n_2\big)\,|n_1n_2\rangle.
\end{aligned}$$

For $\hat{\vec{J}}^2\,|n_1n_2\rangle$, we have

$$\begin{aligned}
\hat{\vec{J}}^2\,|n_1n_2\rangle &= \frac{\hbar^2}{4}\big(\hat{a}_1^\dagger\hat{a}_1 + \hat{a}_2^\dagger\hat{a}_2\big)\big(\hat{a}_1^\dagger\hat{a}_1 + \hat{a}_2^\dagger\hat{a}_2 + 2\big)\,|n_1n_2\rangle \\
&= \frac{\hbar^2}{4}\big(n_1 + n_2\big)\big(n_1 + n_2 + 2\big)\,|n_1n_2\rangle.
\end{aligned}$$

Making use of

$$\hat{J}_1 + i\hat{J}_2 = \hbar\hat{a}_1^\dagger\hat{a}_2,$$

we have

$$\big(\hat{J}_1 + i\hat{J}_2\big)\,|n_1n_2\rangle = \hbar\hat{a}_1^\dagger\hat{a}_2\,|n_1n_2\rangle = \hbar\sqrt{(n_1+1)n_2}\,|n_1+1, n_2-1\rangle.$$

Making use of

$$\hat{J}_1 - i\hat{J}_2 = \hbar\hat{a}_2^\dagger\hat{a}_1,$$

we have

$$\big(\hat{J}_1 - i\hat{J}_2\big)\,|n_1n_2\rangle = \hbar\hat{a}_2^\dagger\hat{a}_1\,|n_1n_2\rangle = \hbar\sqrt{(n_2+1)n_1}\,|n_1-1, n_2+1\rangle.$$

From $\hat{J}_3\,|n_1n_2\rangle = \frac{\hbar}{2}\big(n_1 - n_2\big)\,|n_1n_2\rangle$ and $\hat{\vec{J}}^2\,|n_1n_2\rangle = \frac{\hbar^2}{4}\big(n_1 + n_2\big)\big(n_1 + n_2 + 2\big)\,|n_1n_2\rangle$, we may set

$$\begin{aligned}
j &= \frac{1}{2}\big(n_1 + n_2\big), \\
m &= \frac{1}{2}\big(n_1 - n_2\big).
\end{aligned}$$

Since n_1 and n_2 take on nonnegative integral values only, we have $-j \le m \le j$ and $j = 0, 1/2, 1, 3/2, 2, \cdots$. Solving for n_1 and n_2 from the above two equations, we obtain

$$\begin{aligned} n_1 &= j + m, \\ n_2 &= j - m. \end{aligned}$$

We then have

$$\begin{aligned} (n_1 + 1)n_2 &= (j - m)(j + m + 1), \\ (n_2 + 1)n_1 &= (j + m)(j - m + 1). \end{aligned}$$

Let $|n_1 n_2\rangle$ be denoted by $|jm\rangle$. Since

$$\begin{aligned} \frac{1}{2}\big[(n_1 + 1) + (n_2 - 1)\big] &= \frac{1}{2}\big(n_1 + n_2\big) = j, \\ \frac{1}{2}\big[(n_1 + 1) - (n_2 - 1)\big] &= \frac{1}{2}\big(n_1 - n_2 + 2\big) = m + 1, \end{aligned}$$

we rewrite $|n_1 + 1, n_2 - 1\rangle$ as $|j, m + 1\rangle$. Since

$$\begin{aligned} \frac{1}{2}\big[(n_1 - 1) + (n_2 + 1)\big] &= \frac{1}{2}\big(n_1 + n_2\big) = j, \\ \frac{1}{2}\big[(n_1 - 1) - (n_2 + 1)\big] &= \frac{1}{2}\big(n_1 - n_2 - 2\big) = m - 1, \end{aligned}$$

we rewrite $|n_1 - 1, n_2 + 1\rangle$ as $|j, m - 1\rangle$. Thus, we can rewrite the results for $\hat{J}_3 |n_1 n_2\rangle$, $\hat{\vec{J}}^2 |n_1 n_2\rangle$, and $\big(\hat{J}_1 \pm i\hat{J}_2\big) |n_1 n_2\rangle$ as

$$\begin{aligned} &\hat{J}_3 |jm\rangle = m\hbar |jm\rangle, \\ &\hat{\vec{J}}^2 |jm\rangle = j(j + 1)\hbar^2 |jm\rangle, \\ &\big(\hat{J}_1 \pm i\hat{J}_2\big) |jm\rangle = \hbar\sqrt{(j \mp m)(j \pm m + 1)}\, |j, m \pm 1\rangle . \end{aligned}$$